2-step PCR mixture and conditions (Barcoded-head primers for seqs pooling)

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Abstract

PCR mixture and condition (2X SUPERGREEN PCR MASTER MIX)

Procedure

1 Wear glove, clean up the working bench w. 1% bleach

For 1' PCR head-primers

2 Prepare 1' PCR master mixutre for **head-primers** (**prepare 1.2X of solutions for pipetting error if needed**)
PCR mixture for head-primers for each reaction

Component	Volume	Volume (1.2X)	Final conc.
PowerPol 2X PCR Master Mix	12.5 µL	15 µL	1X
Forward Primer (10 µM)	0.5 µL	1.2 µL	0.2 µM
Reverse Primer (10 µM)	0.5 µL	1.2 µL	0.2 µM
ddH2O	10.5 µL	11.1 µL	-
Total volume	24 µL	28.5 µL	

Component	Volume	Volume (1.2X)	Final conc.
2X KAPA HiFi HotStart ReadyMix	12.5 µL	15 µL	1X
Forward Primer (10 µM)	0.75 µL	0.9 µL	0.3 µM
Forward Primer (10 µM)	0.75 µL	0.9 µL	0.3 µM
ddH2O	10 µL	12 µL	-
Total volume	24 µL	28.8 µL	-

NOTE Negative control ALWAYS NEEDED! For example, if you have 5 PCR reactions to run, prepare master mixture for 6 reactions (5 DNA template + 1 negative control).

3 Mix the 1′ PCR master mixture gently by pippeting. Quick spin the tube.

4 Transfer **24 µL** 1′ PCR master mixutre in 8-strip PCR tubes.

5 Add **1 µL** DNA template in 8-strip PCR tubes, resulting in a **25 µL** reaction mixture for 1′ PCR.

NOTE **Negative control** contains only **24 µL** master mixture but not DNA template

6 Mix the reaction mixture gently by tapping the tubes. Quick spin the tubes.

7 Carry out PCR using the following condition:
1′ PCR condition for **head-primers**

Step	Temp	Sec	Cycle
Initial denaturation	95 °C	180	
Denaturation	95 °C	30	25 cycles
Annealing	60-66 °C varied (b)	30	
Extension	72 °C	180	
Final extension	72 °C	420	
Preservation	Preservation	4 °C	∞

b. Annealing varied, 60-66C is working; Refer to 1′ PCR primers for annealing temperature
c. 1kb ~ 1min extension; enough time allow full extension of sequence

8 Carry out **electrophoresis** for inspection of DNA products

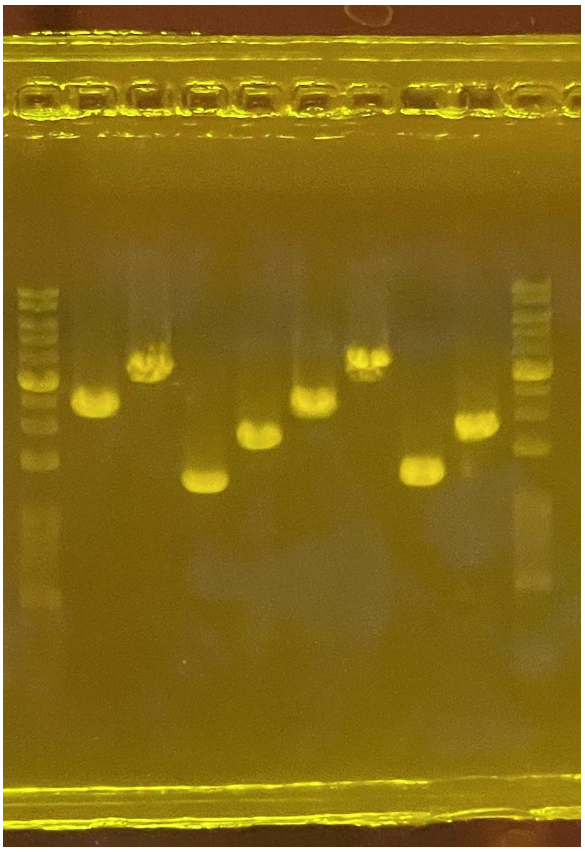


Figure 1: Gel before markdown

9 Markdown wells and upload the pictures to the Lab Google drive

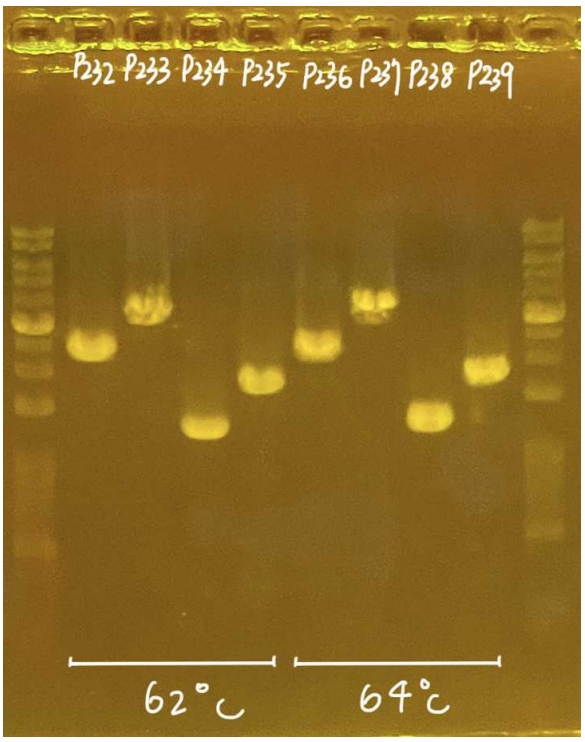


Figure 2: Marked gel picture go to the Lab Google drive

For 2′ PCR barcoded-head primers

10 Prepare 2′ PCR master mixutre for **barcoded-primers** (**prepare 1.2X of solutions for pipetting error if needed**)
PCR mixture for barcoded-primers for each reaction **(NO PRIMERs at this point!!)**

Component	Volume	Volume (1.2X)	Final conc.
ZEJU PCR Master Mix	7.5 µL	9 µL	-
ddH20	5.55 µL	6.66 µL	-
Total volume	13.05 µL	15.66 µL	-

NOTE Negative control ALWAYS NEEDED! For example, if you have 5 PCR reactions to run, prepare master mixture for 6 reactions (5 DNA template + 1 negative control).

11 Mix the 2′ PCR master mixture gently by pippeting. Quick spin the tube.

12 Transfer **13.05 µL** of the 2′ PCR master mixture to 8-strip PCR tubes.

13 Add **1.2 µL pre-mixed barcoded-head primers** (Forward + Reverse) to each PCR tubes.

14 Add **0.75 µL** of **1′ PCR product as template**, resulting in **15 µL** reaction mixture for 2′ PCR. **Negative control** contains only **14.25 µL** master mixture and premixed barcoded-head primers but not DNA template

15 Mix gently by tapping the tubes. Quick spin the tubes.

16 Carry out 2′ PCR using the following condition:
2′ PCR condition for **barcoded-head primers**

Step	Temp	Sec	Cycle
Initial denaturation	98 °C	30	
Denaturation	98 °C	15	12 cycles
Annealing	64-68 °C varied (a)	15	
Extension	72 °C	20 (b)	
Final extension	72 °C	210	
Preservation	Preservation	4 °C	∞

a. Annealing varied, 65 C is working based on test on 220531; Refer 2′ PCR primers for annealing temperature b. 1kb ~ 1min extension; enough time allow full extension of sequence

17 Carry out **electrophoresis** for inspection of DNA products

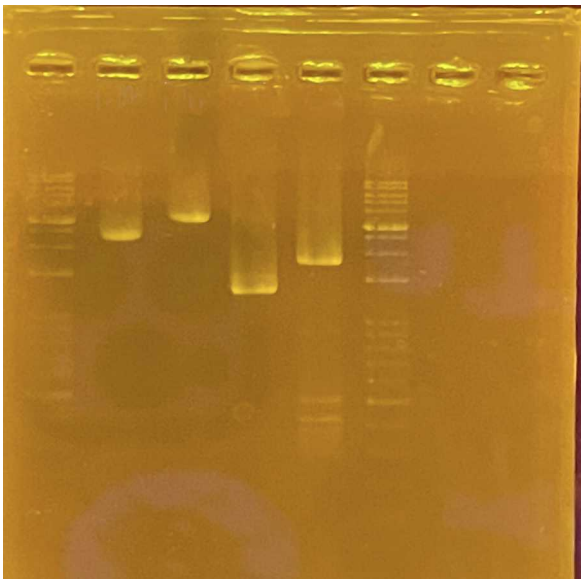


Figure 3: Gel before markdown

18 Markdown wells and upload the pictures to the Lab Google drive

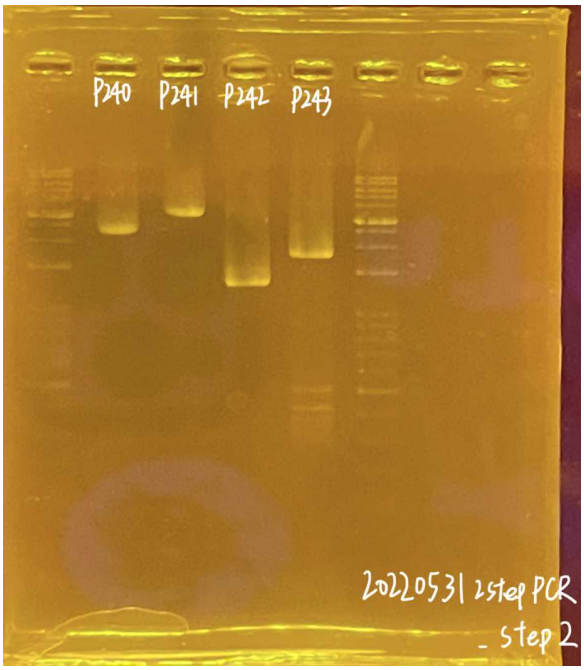


Figure 4: Marked gel picture go to the Lab Google drive